

Coherence Without Content

A missing control for emergent value systems in LLMs

Nullcard

Apart Research Digital Minds Sprint

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The one-slide version

A published metric says these models have coherent values.

We ran it on outcomes that **mean nothing**.

real outcomes	0.906
meaningless outcomes	0.880
difference	+0.025

The metric is not broken. It is **unanchored**.

The claim we are testing

Utility Engineering (Mazeika et al., 2025)

- Ask a model to choose between two outcomes, 500 of them
- Fit a Thurstonian utility model to the choices
- Report held-out accuracy as *structural coherence*
- It rises with scale $\Rightarrow$ “value systems emerge”

Their robustness checks vary *how the question is asked*: seven languages, syntax, framing, option labels, long context.

No condition varies whether the outcomes mean anything.

Why that gap is reachable, not pedantic

High held-out accuracy proves choices follow a *stable scalar ordering*. It does not prove the ordering is *about* anything.

- **244 of 510 outcomes contain a numeral** — orderable by arithmetic alone. Whole categories are 100% numeric.
- In simulation, a responder ordering purely by magnitude, with **zero semantics**, scores **0.949–0.977** held out.
- A responder with a genuine latent utility function scores 0.858–0.928 — **lower**.

An instrument where “count the number” beats “have values” cannot tell them apart from its own output.

The control

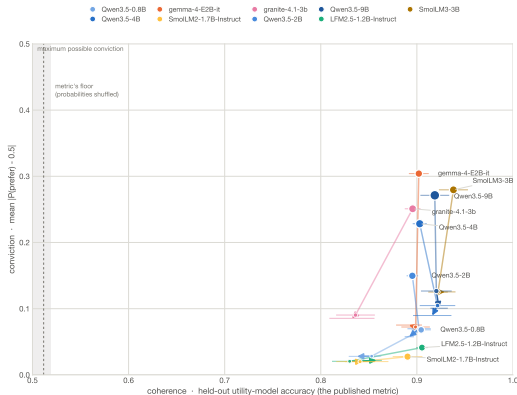
Same 510 outcomes. Same prompt, verbatim. Same fit. Same metric. **Only the referents change.**

arm	outcomes	coherence would reflect
R	real	values + arithmetic + format
N⁺	invented, magnitudes kept	arithmetic + format
N⁻	invented, magnitudes gone	format alone

N⁻ is the floor: what the procedure returns when there is nothing to have a preference *about*.

Every number we report is **R** – **N⁻**, never a raw coherence. 9 models, 81 cells, 3 design replicates each.

Result 1 — the metric barely notices



Each model is a *path*: real $\rightarrow$ invented-with-magnitudes $\rightarrow$ invented. If the metric tracked meaning, paths would run down-*and-left*. They run **almost straight down**.

Result 2 — direction survives, conviction does not

The metric thresholds preferences to hard labels. A pair at $p = 0.51$ counts exactly like a pair at $p = 0.99$.

commits to a side, real outcomes	41%
commits to a side, invented outcomes	4.5%
median collapse	17×

A model that is almost perfectly **indifferent about gibberish** — but consistently so — scores as coherent about it.

Result 3 — the scaling claim, floor-corrected

Pre-registered decisive test. One family, size the only variable, Qwen3.5-0.8B → Qwen3.5-9B.

- If values emerge with scale: floor stays flat, signal rises.
- If it is general answer-consistency: **the floor rises too**.

real-outcome coherence rises by	+0.014
the floor rises by	+0.085
floor-corrected residual	-0.071

Bigger models here are not more coherent *about the outcomes*. They are more consistent in general.

Pooled over all 9 models $r = -0.67$ ($p = 0.043$) — but that is *carried by this family* (within 4 sizes: -0.84; the other 5 models: -0.19). **We claim it within one family only.**

“Maybe your instrument just can’t see anything”

Fair objection. So: install a persona and measure the shift on **both arms**.

$$1 - \frac{\|\Delta_{\text{invented}}\|}{\|\Delta_{\text{real}}\|}$$

1.0 = pure preference change
0.0 = pure style change

- **18 of 20** conditions exceed +0.30, up to +0.85
- One informative exception at -0.66 — moved *gibberish* further than substance. That is what style looks like.
- **Depth barely matters**: user turn vs. system prompt shifts it 0.109; swapping the trait shifts it 0.276

So the instrument *can* see content. It is the unmanipulated coherence number that fails to depend on it.

Three checks that came out in their favour

We report these regardless. They **narrow** our claim.

1. **Order effects are handled.** Pure positional bias produces exactly 50/50 on counterbalanced pairs. It cancels completely.
2. **Not inflated by overfitting.** Held-out evaluation keeps a coin flip at 0.46–0.49. *Our first draft claimed otherwise and was wrong.*
3. **The fit is sound.** Shuffle probabilities across pairs and the metric lands at ~ 0.50 .

We are not saying the metric is broken.

We are saying **a high score does not distinguish values from format.**

What we would want you to hold against us

- **Tokenisation is not matched — so we tested it.** Invented outcomes carry $2.0\times$ the tokens. Matched on tokens (13–24), the residual is $+0.021$ vs $+0.025$ — it **survives**, and the length part runs against us.
- **Is the floor just length?** No — we checked. A one-parameter length rule gets 0.695 on the invented arm (vs 0.541 on real). But with length neutralised the fit still scores **0.848**. Models impose *rich* order on nonsense, not one cheap cue.
- **Scope.** 9 open-weight models, 0.8B–9B, one outcome set. The scaling claim rests on one family.
- **We shipped a bug and corrected it.** Six truncated result files reached an earlier analysis; one was 10% complete. Now every cell is row-count verified at two stages. Headline moved 0.002; one previously reported “instability” was the bug and is withdrawn.

A coherence number without a content control
does not distinguish **having values**
from **being consistent**.

The control costs one battery and a few dollars of GPU.
Reading the number without it costs more.

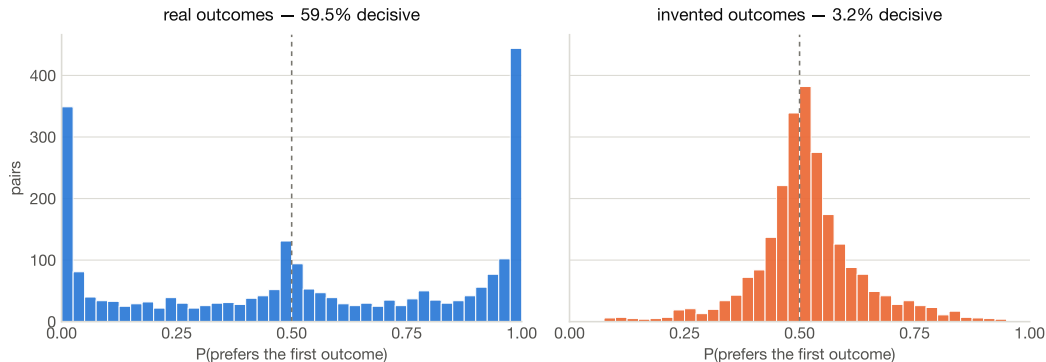
Pre-registered before the first run · battery hash-pinned 342db046213099ad... · code,
pre-registration and figures released.

Backup — all numbers

model	R	N ⁻	R-N ⁻	floor	clears	dec. R	dec. N ⁻
HuggingFaceTB/SmolLM3-3B	0.938	0.929	+0.009	0.029	no	48.7%	4.6%
Qwen/Qwen3.5-9B	0.919	0.923	-0.004	0.010	no	45.7%	2.7%
LiquidAI/LFM2.5-1.2B-Instruct	0.905	0.858	+0.047	0.023	2.0×	0.1%	0.0%
Qwen/Qwen3.5-0.8B	0.904	0.838	+0.067	0.023	2.9×	0.1%	0.0%
Qwen/Qwen3.5-4B	0.903	0.916	-0.013	0.015	no	33.8%	1.9%
google/gemma-4-E2B-it	0.902	0.892	+0.010	0.007	1.5×	56.4%	2.8%
ibm-granite/granite-4.1-3b	0.896	0.832	+0.063	0.029	2.2×	49.7%	14.9%
Qwen/Qwen3.5-2B	0.895	0.892	+0.003	0.001	> floor*	12.8%	0.0%
HuggingFaceTB/SmolLM2-1.7B-Instruct	0.890	0.843	+0.047	0.015	3.0×	0.0%	0.0%

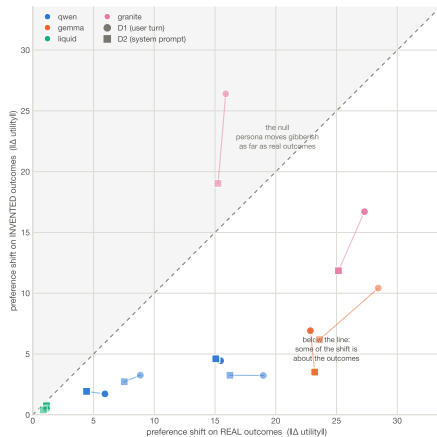
floor = range of the same cell across three independent designs. * = floor too near zero for a ratio to mean anything.

Backup — the mechanism, directly



Real outcomes: mass at the edges. Invented: mass at indifference. Coherence reads only *which side of 0.5*.

Backup — persona displacement



Dashed diagonal is the null. **Below** it = displacement the invented-outcome control cannot explain.